Journal of Strength and Conditioning Research Publish Ahead of Print
DOI: 10.1519/JSC.0000000000002165

ISOMETRIC MID-THIGH PULL PERFORMANCE IS ASSOCIATED WITH ATHLETIC PERFORMANCE AND SPRINTING KINETICS IN DIVISION I MEN AND WOMEN'S BASKETBALL PLAYERS

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Running Head: IMTP and Sprint Performance

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ABSTRACT

PURPOSE: To examine the relationships between isometric mid-thigh pull (IMTP) force, athletic performance measures, and sprint kinetics in Division I men's and women's basketball players. **METHODS:** Twenty-three (male = 8, female = 15) division 1 basketball players completed a maximal 20-m sprint trial while tethered to a device which provided kinetic feedback (peak and average sprinting power, velocity and force). Additionally, one repetition-maximal (1RM) front squat, 1RM hang clean, vertical jump height, and agility (pro-agility and lane agility) tests were performed. Rate of force development (RFD) at 50ms, 100ms, 150ms, 200ms and 250ms of IMTP, as well as peak force (PF) were also collected. Pearson product-moment correlation analysis was used to examine the relationships between these measures. **RESULTS:** Significant ($p < 0.05$) relationships were observed between IMTP PF and sprint time over all distances (5 – 20m; r = –0.62 to 0.69), average sprint velocity (r = 0.50 to 0.70), peak sprint velocity (r = 0.50 to 0.54), average sprint force (r = 0.48 to 0.69), and average sprint power (r = 0.62 to 0.73). Sprinting kinetic measures (average force and power) over the first 5 meters were also significantly ($p < 0.05$) related to IMTP RFD (50 – 250ms; r = 0.42 to 0.62). **CONCLUSION:** Results indicate that IMTP variables are significantly associated with 20-m sprint kinetics. Specifically, IMTP RFD appears to be related to the initial acceleration kinetics of a sprint. Strength and conditioning professionals can possibly implement the IMTP for improved assessment and monitoring of athletic performance and training.

KEYWORDS: Sprinting kinetics; Basketball players; Rate of Force Development; Assessment

INTRODUCTION

Basketball is a team sport that requires players to perform frequent jumps and sprints of varying intensity over the course of a competitive game (25, 31). Furthermore, these movements are often made more complex as they are performed in multiple directions (e.g., forward sprinting, back pedaling, shuffling, and cutting) and may be coordinated with sports-specific skills such as dribbling, passing, or shooting (5). To perform these tasks, basketball players depend heavily on muscular power, strength, speed, and agility (5, 10). Consequently, coaches and trainers report extensive use of plyometric, sprints, and Olympic-style lifts to train and test these performance measures (33). In support, evidence suggests that players who possess greater leg strength, vertical jumping ability, sprinting speed, and agility are awarded more playing time in Division I basketball play (18). While being proficient in these skills has been established across a variety of sports (23, 37, 40), it has been suggested that an athlete's rate of force development (RFD), or ability to reach high force outputs quickly, may be a superior indicator of sport performance (14, 20, 29). Therefore, the implementation of additional testing methods which assess RFD may provide additional insight into the physiological profile of the basketball athlete, as well as a more precise means of tracking adaptations to training.

The isometric mid-thigh pull (IMTP) is a time efficient laboratory-based test designed to reliably and accurately assess peak force (PF) production and rate of force development (RFD) across various time domains (11). Values obtained from the IMTP test have been shown to correlate well to athletic abilities such as speed, agility, weightlifting and vertical jump along with sport-specific performance (3, 23, 24, 37, 40). Due to the ease of its administration and its application to performance in a variety of sports, it is possible that the IMTP test could be used to make inferences on basketball athletic ability in lieu or in addition to a full testing battery.

Sprinting ability is thought to be an important factor in basketball performance (2, 5, 10, 18) and is commonly assessed by the ¾ court (20m) sprint at the professional level (i.e., the National Basketball Association) (26, 33). However, a reoccurring obstacle in sprint analysis is the limitation of sprint time as the sole or primary outcome variable. During game play, linear sprinting is often impaired by the need to avoid other players, contest opposing players, and/or perform sports-specific tasks with the basketball. Thus, assessing factors that contribute to sprint time (i.e., force, velocity, and power) may be more useful for assessing basketball athletes. While examining these kinetic factors in basketball athletes is intriguing, currently no study has determined their relationship to traditional measures of athletic performance.

Therefore the purpose of this study was to detect associations between isometric force production and basketball-relevant performance measures in Division I Men and Women's basketball athletes. Additionally, we sought to determine if any relationships exist between IMTP performance and measures of sprinting power, force and speed. Practically, the purpose of this study was to provide additional insight to the physiological profile of Division I basketball players, as well as identify potentially beneficial performance assessments to aid in the evaluation of athletic development.

METHODS

Experimental Approach to the Problem

The relationship between isometric force production, athletic performance and sprinting kinetics was examined in twenty-three male and female collegiate basketball players. All participants reported as a team to their normal practice gymnasium during the summer prior to the 2016-17 regular season. Following an explanation of the purpose, risks, and benefits associated with the study, the participants completed a standardized warm-up, IMTP testing, and maximal sprint trials tethered to a measurement device. All sprint and IMTP trials were verbally encouraged and conducted in the presence of their teammates. Maximal strength (front squat, hang clean) and performance testing (vertical jump, pro-agility, lane agility) were administered on a separate day by the teams' Certified Strength and Conditioning Specialist.

Participants

Twenty-three Division I male (n = 8, 20.0 ± 1.7 y, 1.9 ± 0.12 m, 87.0 ± 13.9 kg) and female (n = 15, 19.9 ± 1.1 y, 1.76 ± 0.10 m, 72.7 ± 12.3 kg) basketball players volunteered to participate in this investigation. Prior to participation, each athlete was asked to complete a medical and activity questionnaire, a physical activity readiness questionnaire (PAR-Q), and provide their (or their legal guardian's) written informed consent following explanation. All participants were free of any physical limitations (determined by medical and activity questionnaire) and had been actively participating in the same off-season training program. This research protocol was approved by the Institutional Review Board of the University. Following an explanation of the purpose, risks, and benefits associated with the study, the athletes completed a standardized warm-up and then a maximal sprint trial. In an attempt to eliminate the potential for reduced performance, the participants were asked to refrain from any strenuous physical activity for the previous 48 hours.

Maximal Strength Testing

Front Squat Exercise. Maximal strength testing was performed on the front squat exercise. All 1RM testing was performed using methods previously described (17). Prior to beginning the test, each athlete completed a general warm-up led by the strength and conditioning coach which included jogging and a dynamic warm-up. Each athlete performed a warm-up set using a resistance that was approximately 40–60% of their perceived maximum and then performed 3–4 maximal, single-repetition trials to determine the 1RM. A 3- to 5-min rest period was provided between each trial. The squat exercise required each athlete to descend to the parallel position, which was attained when the greater trochanter of the femur reached the same level as the knee. The athlete then ascended until full knee extension. Maximum strength testing was administered by the same Certified Strength and Conditioning Specialist to ensure that each participant reached the parallel position for each repetition of the squat and that the exercise technique was consistent between sessions.

1RM Hang Clean. Hang power clean began from a position such that the athlete stood in an athletic stance and held the barbell in front of their body at the mid-thigh position. The athlete started the movement by lowering the barbell to a position just above their knee and then explosively moving the barbell upward to be received at shoulder height. The strength and conditioning coach estimated the athlete's starting warm-up weight from their recent training log, and planned the weights to be lifted during a series of approximately 2-3 warm-up sets. In each set, the athlete performed one to three repetitions, and the weight was increased every set. The athlete started the warm-up set with the bar only (20 kg) and then added 20–40 kg on each set until the load was about 60% of estimated 1RM. Subsequently, loads were increased by 5–10 kg each set until the load was 90% of estimated 1RM. After these sets were completed, the weight was increased by 2.5 or 5 kg after each successful attempt until their 1RM was determined. A rest period of at least 3-minutes was given between each 1RM attempt.

Isometric Mid-thigh Pull Testing. The mid-thigh position was determined for each athlete before testing by marking the midpoint distance between the knee and hip joints. Each athlete was instructed to assume their preferred second pull power-clean position by self-selecting their hip and knee angles. The height of the barbell was then adjusted up or down to make sure it was placed at the mid-thigh. All athletes used an overhand grip and were strapped to the barbell in a manner similar to previous investigations (11, 13). The athletes were instructed to pull upwards on the barbell as hard and as fast as possible and to continue their maximal effort for 6 seconds. All athletes were instructed to relax before the command "GO!" to avoid precontraction. The force-time curve for each trial was recorded by dual force plates (PASCO, Roseville, CA, USA) with a sample rate of 1,000 Hz similarly to previous investigations (11, 24, 40).

Peak force was defined as the highest force achieved during the 6-second isometric test minus the participant's body weight in Newtons. Additionally, force outputs at 50, 100, 150, 200 and 250ms from the initiation of the pull were determined for each trial. The RFD was then calculated using the following equation: RFD = ΔForce / ΔTime. The RFD equation was applied to predetermined time bands: 0-50ms, 0-100ms, 0-150ms, 0-200ms and 0-250ms. This was in accordance with previous studies that used similar predetermined time bands when calculating force and RFD and demonstrating high reliability (11, 15, 22, 24, 41). These time intervals were selected based on typical ground-contact times experienced during various athletic movements (>50ms) for example, sprinting, jumping, and changing direction (19, 32, 42).

Performance Testing

Vertical Jump. A Vertical Jump testing station (Uesaka Sport, Colorado Springs, CO) was used to assess vertical jump height (± 0. 5 in). Prior to the test, each athlete's standing vertical reach height was determined by colored squares located along the vertical neck of the device. These squares correspond with similarly colored markings on each horizontal tab, which indicate the vertical distance from the associated square. Vertical jump height was determined by the indicated distance on the highest tab reached following 3 maximal countermovement jump (CMJ) attempts performed from a standing position with feet shoulder width apart.

Pro-Agility Testing. For the pro-agility test, three cones were placed parallel, five meters apart. The athletes set up for the test in a straddle position facing the middle cone. On their ready, the athletes were instructed to pivot and accelerate as quickly as possible to a cone 5m away and then upon reaching the first cone, pivot again and sprint the 10m distance to the furthest cone. Upon reaching this cone, the athletes once again pivoted to return to the middle cone as quickly as possible. During each change in direction, the athletes were asked to touch the ground next to the cone. Trials where the athlete failed to touch the ground were discarded. Athletes were allowed three attempts and the fastest time measured in seconds was recorded.

Lane agility Testing. For the lane agility test, cones were placed at the 4 corners of the key on a basketball court (2 at the right and left corners of the free throw line and 2 at the right and left hand corners of the baseline). Standing at the left hand corner of the free throw line facing the baseline, the athletes would sprint forward to the first cone at the baseline, side shuffle right to the second cone at the baseline, backpedal to the third cone at the free throw line, side shuffle left to the fourth cone at the free throw line (at the start position), change directions to shuffle to the right back to the third cone, sprint forward to the second cone, side shuffle left to the first cone, and finish by backpedaling to the fourth cone at the original start position. Time was measured from the individual's first movement to when they completed the drill. Athletes were allowed three attempts and the fastest time measured in seconds was recorded.

Sprint Testing. Sprinting kinetics were measured via the 1080 Sprint (1080 Motion; Lidingö, Sweden). The 1080 Sprint is a portable, cable resistance device that uses a servo motor (2000 RPM OMRON G5 Series Motor, OMRON Corporation, Kyoto, Japan) to modulate resistance load (1 – 15 kg). The motor is connected to a composite fiber cord that is wrapped around a spool and can extend up to 90 meters. The cord was attached to the athlete via shoulder harness (Power Systems LLC, Knoxville, TN). The resistance load was controlled by the Quantum computer application (1080 Motion, Lidingö, Sweden) where all sprinting kinetics data were stored for analysis.

Prior to physical activity, height (± 0.01 m) and weight (± 0.01 kg) were recorded in all participants via a commercial scale and stadiometer (Health-o-meter, Bridgeview, IL, USA). The athletes then completed a standardized general and dynamic warm-up that was consistent with their normal training habits and led by each teams' strength and conditioning coach. Subsequently, the athletes completed one maximal 20-m sprint free of any resistance followed by a maximal 20-m sprint while tethered to the 1080 Sprint (see Figure 1). All sprint tests using the 1080 Sprint were conducted in the "Isotonic" mode to provide the smoothest resistance during testing. The sprint was performed against the minimal resistance allowed by the device (1 kg), which is necessary for kinetic data output. A pair of cones and tape affixed to the floor were positioned approximately 5 meters from the 1080 Sprint to denote the "starting line". The athletes were instructed to take their preferred starting stance at the starting line and to begin each maximal trial at their ready. The Quantum software can detect the athlete's position and then be set to initiate data collection on his/her first movement for a specified distance (i.e., 20 meters). The starting leg was recorded during each sprint. Peak and average sprinting force (N), velocity ($m \cdot sec^{-1}$) and power (Watts) were measured for each sprint from 0 – 5 m, 0 – 10 m, 0 – 15 m, and 0 – 20 m.

[INSERT FIGURE 1 HERE]

Statistical Analysis

Data from this investigation were reported as means ± standard deviations and analyzed using SPSS v24.0 software (SPSS Inc., Chicago, IL). Shapiro-Wilk tests were used to verify normal distribution of data. Relationships between variables were analyzed using a Pearson product moment correlation. Correlation coefficients were evaluated using the following criteria: small = 0.1 to 0.29, moderate = 0.30 to 0.49, large = 0.50 to 0.69, very large = 0.70 to 0.89, nearly perfect = 0.90 to 0.99, and perfect = 1.0 (17). An alpha level of $p \leq 0.05$ was considered statistically significant for all comparisons.

RESULTS

Descriptive statistics for measures of athletic performance (1RM squat, 1RM hang clean, vertical jump, pro-agility, lane-agility) and IMTP results are presented in Table 1.

[INSERT TABLE 1 HERE]

Relationship between IMTP and measures of athletic performance

The relationships between the IMTP and athletic performance measures are presented in Table 2. Briefly, IMTP PF was positively related ($p < 0.01$) to measures of muscular strength and power (r = 0. 71 to 0.89) and inversely related ($p < 0.05$) to agility (r = -0.52 to 0.66). Strong correlations (r = 0.57 to 0.70) were also observed between IMTP RFD and 1RM hang clean (r = 0.67 to 0.70, p < 0.01) and vertical jump (r = 0.56 to 0.57, p < 0.05).

[INERT TABLE 2 HERE]

Relationship of IMTP and Sprint Kinetics

The results of the analyses between the sprint kinetics and IMTP values are presented in Table 3. Briefly, large-to-very large inverse relationships (r = –0.62 to –0.69, p < 0.01) were observed between IMTP Peak force and sprint times over all distances (5-20m). Significant relationships (p < 0.001) were observed between IMTP peak and average power (Figure 2A) and force (Figure 2B) produced across the entire 20-m sprint. Furthermore, consistent moderate-to-large positive relationships (r = 0.42 to 0.62, p < 0.05) were observed between IMTP RFD at each time band (50, 100, 150, 200, 250ms) and average power (Figure 2C) and force (Figure 2D) over 5 meters.

[INSERT TABLE 3 HERE]

DISCUSSION

The primary purpose of this study was to examine the relationships between isometric strength measures (i.e., peak force production and RFD), dynamic measures of strength (1RM Front Squat), power (1RM Hang Clean and vertical jump), agility, and 20-m sprinting kinetics in Division I basketball athletes. Our data suggests that strong relationships exist between peak isometric force and 20-m sprint time, dynamic strength and power, and agility. Furthermore, peak isometric force was also related to the average power and force produced during a 20-m sprint, while IMTP RFD was related to sprinting kinetics (average power and force) over the first

5m of the maximal sprint. To our knowledge, this is the first study to assess the relationships between variables obtained from an IMTP test and sprinting kinetics.

The relationships observed between peak force and measures of strength and power, particularly between IMTP PF and 1RM hang clean ($r = 0.89$), are not surprising. The IMTP test is designed to imitate the body's position at the beginning of the second pull of the clean exercise. Several previous investigations have reported strong relationships between maximal isometric strength and dynamic lifts such as the snatch and clean & jerk (3, 13, 36) and vertical jump performance (static and countermovement) (21, 22). Similarly, we observed relationships between IMTP RFD and measures of power (1RM hang clean and vertical jump), though only at 200 – 250 ms. Kraska and colleagues observed significant correlations between isometric RFD and vertical jump height in Division I athletes which were similar to correlations we found with IMTP RFD (22). Our findings support the notion that better jumpers possess a high capacity for contractile strength and explosiveness (21, 36, 37) and are thus, more likely to be awarded more playing time (18). Furthermore, the vertical jump test has been reported to be the most frequently implemented athletic performance test among NBA strength and conditioning coaches along with other evaluations of muscular power (high pulls, cleans) (33). However, these individual assessments do not provide additional insight into the athlete's capabilities beyond their intended measure (i.e., jumping height and weight lifted). Considering foot contact times of 250-300ms are experienced during vertical jump performance (30), the IMTP may be a more comprehensive alternative assessment of strength and explosiveness.

Peak force from the IMTP was inversely related to pro-agility and lane agility times which is in agreement with previous reports (34, 38-40). However, we observed no correlation between the IMTP RFD at any time band which is in contrast to previous reports of relationships peak IMTP RFD and performance in the 505 agility (7, 38, 39) and the pro-agility (40) tests. In the current study, IMTP PF was determined as the highest force output achieved in a 6-second effort while RFD was only assessed in sampling windows up to 250ms. Sasaki et al.(32) previously reported mean ground contact times of 440ms during change of direction performance tasks in male collegiate soccer players. Thus, it is likely that the lack of relationships between RFD and agility we observed can be attributed a discrepancy between RFD sampling windows and typical ground contact times during agility tasks. While the pro-agility test is a commonly implemented in basketball testing, it likely does not mimic the exact change of direction movements that basketball athletes experience during gameplay (e.g. shuffling, side stepping). Additional disagreements with the literature may be attributed to the complex organization of movement and sport-specific change of directions that occur specifically in the lane-agility drill compared to assessments utilized in previous studies.

Time-motion analysis has revealed that basketball players spend a high percentage of live-game playing time performing high-intensity sport-specific movements and sprints (4). With our athletes, we detected large to very large inverse relationships between IMTP PF and sprint time at all distances (5-, 10-, 15-, 20-m). These findings are consistent with previous reports (7, 39, 41), specifically those examining relationships between IMTP peak force and sprint times at 5- (r = -0.58), 10 (r = 0.61), and 20-m (r = 0.62) (7). Interestingly, IMTP RFD only related to sprint times (> 10 m) at 200 and 250ms which is in contrast to those reported by Wang et al. (2017). In that study, RFD at 30 – 50ms was only related to sprint time over the initial 5m (r = -0.52 to -0.57) of a 40-m sprint (40). It is possible that the slower 20m sprint times recorded in the present study (3.61s) compared to Wang et al. (3.06s) contributed to our different findings as faster athletes have been shown to have shorter mean ground contact times (126ms) compared their slower counterparts (180-210ms)(6, 28). Our slower reported times may be attributed to the inclusion of male and female athletes into our analysis, and that the basketball athletes sprinted against minimal resistance (1kg) while tethered to the 1080 device. Practically speaking, IMTP RFD time band and PF data likely should be interpreted by taking typical sport-specific ground contact times in consideration when making inferences to athletic performance. Moreover, due to the large associations between PF and sprint performance, the development and testing of strength and explosiveness should remain a primary goal in the training of basketball athletes.

While short-distance sprint time is an important indicator of an athlete's ability to accelerate, the context of the test (i.e., it is performed without opposition or need to modify direction) may not be specific to acceleration requirements during game play. Consequently, the kinetic factors (i.e., force, velocity, and power) produced during a sprint test may be more indicative of an athlete's capabilities in a competitive setting. In the present study, moderate to large relationships were observed between RFD (50 – 250ms) and 5-m sprinting kinetics (average power and force), as well as between RFD (100 – 250ms) and 20-m sprinting kinetics (average force, power, and velocity). The general absence of statistically significant relationships between IMTP RFD and sprinting kinetic measures at 10- and 15-m is difficult to explain. Previous literature suggests that the acceleration phase of a sprint involves a greater amount of knee extensor (vastus lateralis and medialis) activity and activation compared to activity observed at maximal velocity (16, 27, 43). Given that high amplitudes of vastus lateralis activation have been reported during Olympic style lifting (12), it is likely we found the greatest agreement between IMTP RFD and 5m sprint kinetics due to the shared reliance of musculature between the activities. Taken together, it appears that IMTP RFD is best related to sprint take-off power and force production, an essential attribute in the sport of basketball which requires efficient repeated short sprints for optimal sport performance (2, 5, 10, 35). Future investigations examining the differences in electromyographic (EMG) activation patterns is needed to make any additional conclusions between the IMTP and 20-m sprint.

It is also important to note that optimal sprinting performance requires the summation of large propulsion forces in both the vertical and horizontal planes (1, 8, 9). In the present study, the measurement device (1080 Sprint) requires a minimal resistance (1 kg) as a default setting for kinetic measurement and this force is applied predominantly in the horizontal plane similar to a sled towing device (8). Therefore, kinetic data obtained from the maximal sprint was representative of primarily horizontal-vector kinetics. Due to the fact that the IMTP measures peak force and RFD solely in the vertical plane, it is not surprising that perfect correlations between IMTP and sprint kinetics were not present in our study. Nevertheless, the 1080 Sprint provides new possibilities for future endeavors in sport and movement analytics. Though the 1080 Sprint is marketed to consumers as a means of providing real-time data during various activities (e.g. sprinting, skating, swimming, or change of direction), more work is needed to validate the 1080 Sprint as a data collection tool across various movement patterns. Additionally, while all athletes were trained and proficient in the hang clean, the IMTP was a novel test for these participants. While familiarization and practice was conducted before their performance was recorded and used in analysis, this may have affected RFD and correlational data. Further work is needed to determine if relationships between the IMTP and sprint performance are altered as athletes continue to become accustomed to testing procedures. This would provide useful information for scientists and practitioners when tracking changes in athletic performance.

Practical Applications

Strength and conditioning professionals, coaches and sport scientists strive to utilize empirical, sport-specific assessment protocols and technologies for optimal performance. The results of this study demonstrate relationships between several measures of strength, power, speed, and agility that are important for collegiate basketball players. In particular, the IMTP test alone provides information about an athlete's strength and explosiveness which appear to be related to several individual assessments of these attributes. Moreover, the relationships observed between IMTP measures (peak force production and RFD) and sprinting kinetics during the acceleration phase of a sprint (i.e., the first 5m) and over the entire 20-m sprint, may provide more insight into an athlete's capabilities during actual game play. Consequently, basketball coaches, trainers and athletes might consider implementing the IMTP test in lieu of (or in addition to) a full testing battery, for athletic evaluation and monitoring during training or a competitive season.

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FIGURE LEGEND

Figure 1. Collegiate athlete (A) tethered to the 1080 Sprinting Device (B).

Figure 2. Scatter plot correlative findings between (A) IMTP peak force and average power during 20m Sprint (B) IMTP peak force and average force during 20m sprint (C) RFD 0-200ms and average power during first 5m of sprint (D) RFD 0-200ms and average force during first 5m of sprint. RFD = rate of force development; IMTP = isometric mid-thigh pull. Black marker = female athletes. White marker = male athletes.

Table 1. Descriptive statistics for performance and IMTP results.

	Men's Basketball Team (n=8)	Women's Basketball Team (n=15)
Front Squat 1RM (kg)	126.1 ± 17.7	83.6 ± 12.5
Hang Clean 1RM (kg)	44.0 ± 6.1	23.7 ± 2.7
Vertical Jump (cm)	77.4 ± 6.4	50.0 ± 5.4
Pro-Agility (s)	4.54 ± 0.21	5.05 ± 0.22
Lane Agility (s)	11.80 ± 0.90	13.68 ± 0.86
1080 20m Sprint (s)	3.31 ± 0.15	3.79 ± 0.13
Peak Force (N)	2534.1 ± 368.0	1248.0 ± 377.2
RFD 0-50ms ($N \cdot s^{-1}$)	5643.1 ± 6459.8	2778.7 ± 1708.0
RFD 0-100ms ($N \cdot s^{-1}$)	6161.4 ± 4694.8	2988.1 ± 1831.1
RFD 0-150ms ($N \cdot s^{-1}$)	5811.0 ± 3995.0	2664.6 ± 1439.0
RFD 0-200ms ($N \cdot s^{-1}$)	6015.8 ± 2678.0	2736.0 ± 1176.5
RFD 0-250ms ($N \cdot s^{-1}$)	5603.0 ± 2272.6	2882.2 ± 1105.2

Data presented as mean ± SD

Table 2. Correlations between IMTP and performance measures

	Peak Force (N)	RFD 0-50ms ($N \cdot s^{-1}$)	RFD 0-100ms ($N \cdot s^{-1}$)	RFD 0-150ms ($N \cdot s^{-1}$)	RFD 0-200ms ($N \cdot s^{-1}$)	RFD 0-250ms ($N \cdot s^{-1}$)
Front Squat 1RM(kg)	.705**	.119	.255	.354	.466	.463
Hang Clean 1RM	.890**	.193	.336	.462	.668**	.701**
Vertical Jump (cm)	.809**	.235	.345	.443	.556*	.570*
Pro-Agility (s)	-.657**	-.233	-.341	-.340	-.378	-.358
Lane Agility (s)	-.523*	-.331	- .345	-.340	-.264	-.179

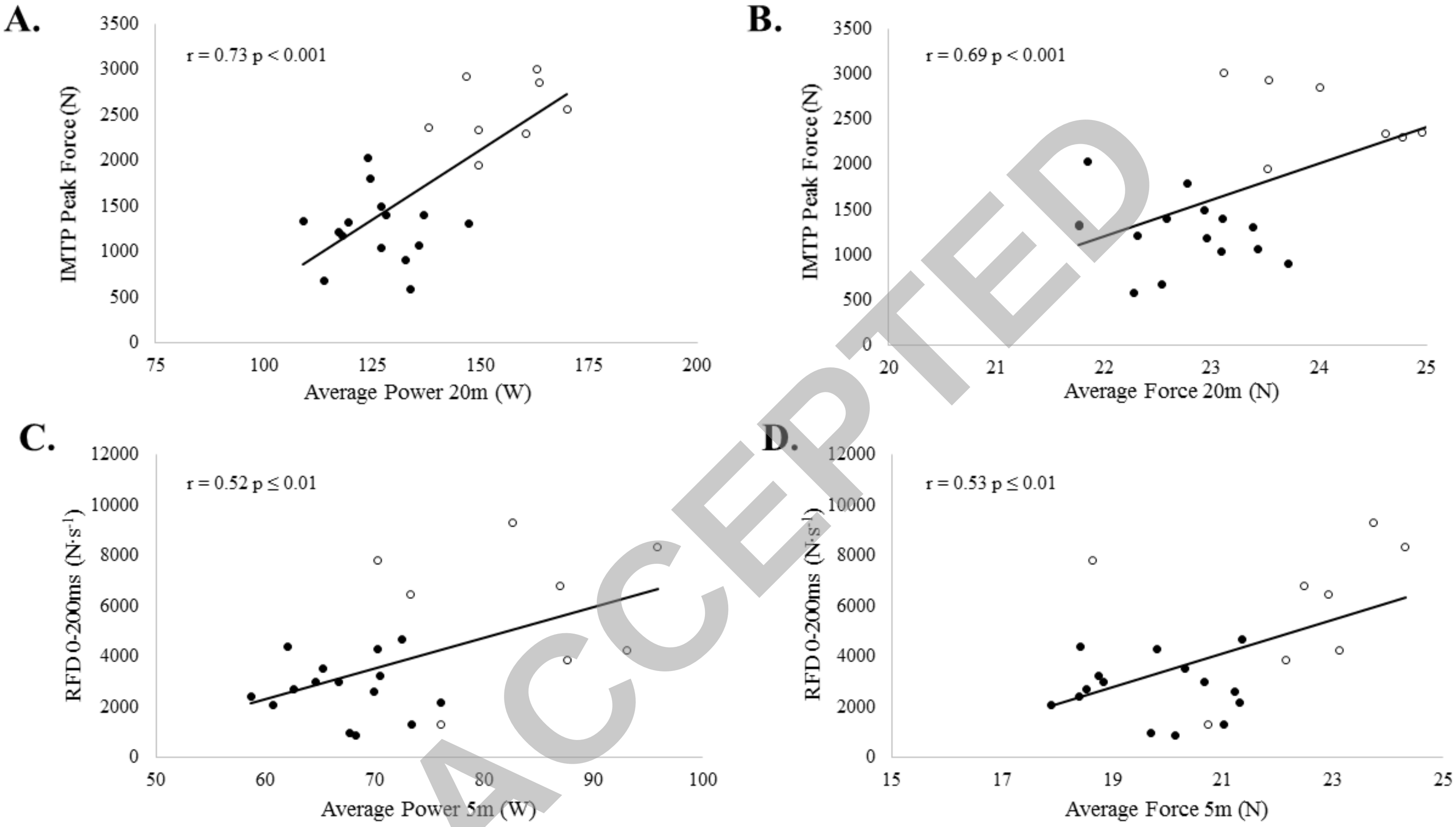
* p ≤ .05, ** p ≤ .01

Table 3. Correlations between IMTP and 20m Sprint Times and Kinetics

		Peak Force (N)	RFD 0-50ms ($N \cdot s^{-1}$)	RFD 0-100 ms ($N \cdot s^{-1}$)	RFD 0-150ms ($N \cdot s^{-1}$)	RFD 0-200ms ($N \cdot s^{-1}$)	RFD 0-250ms ($N \cdot s^{-1}$)
Time (s)	5 meters	-.619**	-.212	-.339	-.319	-.395	-.381
	10 meters	-.673**	-.272	-.373	-.359	-.451*	-.432*
	15 meters	-.696**	-.291	-.387	-.382	-.472*	-.451*
	20 meters	-.693**	-.299	-.386	-.384	-.471*	-.448*
Avg. Velocity ($m \cdot s^{-1}$)	5 meters	.496*	.336	.460	.399	.325	.267
	10 meters	.603**	.409	.486*	.440*	.404	.363
	15 meters	.664**	.317	.426*	.437*	.448*	.403
	20 meters	.704**	.265	.375	.391	.485*	.493*
Peak Velocity ($m \cdot s^{-1}$)	5 meters	.498*	.408	.487*	.455*	.431*	.366
	10 meters	.499*	.282	.326	.381	.480*	.438*
	15 meters	.515*	.283	.300	.275	.345	.325
	20 meters	.536**	.163	.212	.200	.306	.319
Avg. Force (N)	5 meters	.520**	.480*	.535**	.589**	.525**	.415*
	10 meters	.482*	.337	.346	.361	.328	.265
	15 meters	.584*	.301	.347	.369	.356	.311
	20 meters	.685**	.386	.447*	.478*	.515*	.515*
Peak Force (N)	5 meters	.121	.254	.245	.162	.035	.016
	10 meters	.221	-.065	.045	-.051	.010	.034
	15 meters	.207	-.047	-.034	.046	.011	.027
	20 meters	.161	.181	.154	.072	.101	.069
Avg. Power (W)	5 meters	.621**	.504*	.593**	.589**	.519**	.427*
	10 meters	.624**	.418*	.457*	.432*	.400	.351
	15 meters	.686**	.314	.401	.411	.423*	.384
	20 meters	.728**	.312	.408*	.425*	.512*	.523*
Peak Power (W)	5 meters	.186	.309	.332	.240	.130	.070
	10 meters	.337	.066	.091	.054	.105	.116
	15 meters	.292	.065	.066	.019	.074	.073
	20 meters	.379	.173	.187	.156	.244	.233

* $p \leq .05$, ** $p \leq .01$

Figure 1. Collegiate athlete (A) tethered to the 1080 Sprinting Device (B).


A.
r = 0.73 p < 0.001
IMTP Peak Force (N)
Average Power 20m (W)
B.
r = 0.69 p < 0.001
IMTP Peak Force (N)
Average Force 20m (N)
C.
r = 0.52 p ≤ 0.01
RFD 0-200ms ($N \cdot s^{-1}$)
Average Power 5m (W)
D.
r = 0.53 p ≤ 0.01
RFD 0-200ms ($N \cdot s^{-1}$)
Average Force 5m (N)